



TECHNICAL PROFESSIONAL

IA - DRIVING AND AUTOMOTIVE SERVICING

Grade 12

**Three-Term Budget Of Work (BOW)
for Learning Competencies**

TECHNICAL PROFESSIONAL

IA - AUTOMOTIVE AND SMALL ENGINE TECHNOLOGIES

Grade 12

Three-Term Budget Of Work For Learning Competencies

Please note that each TechPro elective is completed within one term if delivered at Grade 12.

Term Plan	
Elective	Driving and Automotive Servicing
Performance Standard	The learners perform procedures for moving and positioning vehicles, operating vehicle lifting and support equipment, and driving light vehicles, employing appropriate safety precautions. The learners perform procedures of pre-delivery inspection in automotive servicing employing appropriate safety precautions. The learners perform procedures of periodic maintenance services (PMS) for vehicle component parts (engine) in automotive servicing, employing appropriate safety precautions. The learners perform automotive periodic maintenance services (PMS) for the vehicle component parts (drive train, brake system, suspension system, and steering system) by employing appropriate safety precautions and conduct vehicle release and service cost calculations.
Content Standard	• Demonstrate an understanding of the concepts and principles of driving a light vehicle. • Demonstrate an understanding of the concepts and principles of pre-delivery inspection (PDI) in automotive servicing. • Demonstrate an understanding of the concepts and principles of periodic maintenance services (PMS) for vehicle component parts (engine) in automotive servicing. • Demonstrate an understanding of the concepts of automotive periodic maintenance services (PMS) for the vehicle's component parts (drive train, brake system, suspension

	system, and steering system).
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Week	Learning Competency	Suggested Activities
1	- Discuss the overview of driving vehicles. - Identify traffic laws/rules and regulations.	Know Your Vehicle - Learners explore the basic parts of a vehicle, dashboard symbols, primary controls (steering wheel, pedals, gear shift), and the fundamental principles of vehicle operation and safe driving. - Identify and explain the function of key vehicle parts and controls using a labeled diagram or actual vehicle demonstration Safe Driving Concepts- Discussion Forum - Learners discuss essential driving concepts such as traffic rules, road signs, driver responsibilities, and basic road safety practices through guided group discussion or case scenarios. - Participate in a group discussion and present key points about safe and responsible driving practices. Traffic Law Discovery Chart - Learners examine common traffic laws, road rules, and driving regulations (e.g., speed limits,

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		right-of-way, seatbelt laws, DUI laws) and organize them into categories such as safety rules, regulatory laws, and driver responsibilities to better understand their purpose and importance. ○ Research and list at least five traffic laws, then classify and explain their purpose in a simple chart. ● Road Rules Scenario Matching Game ○ Learners analyze real-life driving situations and match each scenario to the correct traffic law or regulation, helping them recognize how rules apply in everyday road situations. ○ Match each given driving scenario with the appropriate traffic law or regulation and briefly justify your answer. ● Road Emergency Response Workshop ○ Learners study common road emergencies such as brake failure, tire blowout, engine overheating, minor collisions, and accidents, and discuss the correct step-by-step actions drivers should take to ensure safety and minimize risk. ○ Identify the correct response

Week	Learning Competency	Suggested Activities
		procedures for assigned road emergency situations and present them to the class. • Emergency Scenario Role-Play ○ Learners participate in guided role-play or case simulations where they act out how to properly respond to emergencies, including calling authorities, using hazard lights, setting up warning devices, and providing basic assistance. ○ Demonstrate the appropriate response to a given road emergency scenario through a short role-play presentation.
2	• Demonstrate the role of service personnel in moving and positioning light vehicles.	• Workshop Vehicle Positioning Demonstration ○ Learners observe and practice the correct procedures service personnel follow when moving and positioning light vehicles inside a workshop, including safety checks, use of hand signals, proper use of parking brakes, wheel chocks, and maintaining clear communication. ○ Demonstrate the proper and safe procedure in moving and positioning a light vehicle within a workshop setting.

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		• Service Personnel Role-Play Simulation ○ Learners simulate the roles of service personnel such as spotter, driver, and supervisor to understand teamwork, safety coordination, and proper vehicle handling when positioning vehicles for repair or inspection. ○ Perform a role-play showing the coordinated roles and safety practices of service personnel while moving and positioning a light vehicle.
3	• Identify the type of vehicle and the appropriate maintenance. • According to specifications and power source.	• Vehicle Type and Power Source Classification Task ○ Learners examine different types of vehicles (gasoline, diesel, hybrid, electric) and classify them according to their power source, features, and manufacturer specifications, including identifying the correct basic maintenance requirements for each type. ○ Classify given vehicles by power source and list the recommended maintenance based on their specifications. • Maintenance Matching Workshop ○ Learners analyze sample vehicle

Week	Learning Competency	Suggested Activities
		manuals or specification sheets and match each vehicle type with the appropriate maintenance procedures such as oil change intervals, battery care, coolant checks, and brake inspection. ○ Match each vehicle type with its correct maintenance procedures according to the provided specifications.
4	• Perform the procedures of operating vehicle lifting and support equipment in servicing vehicles.	• Safe Use of Vehicle Lifting Equipment Demonstration ○ Learners are introduced to different vehicle lifting and support equipment such as hydraulic jacks, jack stands, and vehicle lifts. They study proper operating procedures, safety precautions, equipment inspection, correct lifting points, and load capacity to ensure safe servicing practices. ○ Demonstrate the correct and safe procedure for lifting and supporting a vehicle using appropriate equipment. • Lifting Equipment Skills Practice Rotation ○ Learners rotate through stations where they practice positioning a jack,

Week	Learning Competency	Suggested Activities
		placing jack stands, checking stability, and observing proper communication and safety protocols under supervision, reinforcing hands-on competence. ○ Perform the step-by-step process of operating vehicle lifting and support equipment following standard safety procedures.
5	● Perform driving procedures for light vehicles.	● Basic Driving Procedures Practical Drill ○ Learners practice the fundamental driving procedures for light vehicles, including pre-operation checks, proper seat and mirror adjustment, starting and stopping the engine, shifting gears (manual/automatic), steering control, signaling, and safe braking in a controlled environment. ○ Demonstrate the correct step-by-step driving procedures for operating a light vehicle safely in a controlled area. ● Controlled Driving Simulation and Evaluation ○ Learners perform structured driving tasks such as proper lane positioning, turning, parking, and observing traffic signs within a simulated or supervised

Week	Learning Competency	Suggested Activities
		driving course while applying safety rules and correct driving techniques. Perform assigned driving tasks following standard procedures and safety regulations under instructor supervision.
6	- Discuss the pre-delivery inspection (PDI) in automotive servicing. - Identify required items for pre-delivery inspection (PDI) service.	PDI Walkthrough Simulation - This is a hands-on simulation where learners perform a mock Pre-Delivery Inspection on a vehicle (or training unit) using a standard PDI checklist. Learners examine vehicle components such as fluid levels, tire pressure, lights, battery condition, and safety features while understanding the purpose of each inspection item before a vehicle is released to a customer. - Using a standard PDI checklist, inspect the assigned vehicle and explain the purpose of each inspection item as you complete it. PDI Process Flowchart and Group Discussion - Learners work in small groups to create a flowchart or diagram that outlines the complete Pre-Delivery Inspection process—from vehicle

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		arrival to final release—followed by a guided discussion on why PDI is essential for safety, quality control, and customer satisfaction. ○ Create a flowchart of the Pre-Delivery Inspection process and present to the class explaining why each step is important. ● PDI Checklist Identification Drill ○ Learners examine a standard Pre-Delivery Inspection (PDI) checklist and identify the required inspection items under categories such as engine compartment, under chassis, electrical system, interior, and exterior components. Learners classify and explain why each item must be checked before vehicle release. ○ Review the given PDI checklist and highlight or list all required inspection items under their proper categories. ● Spot the Missing Item - Workshop Activity ○ Learners are presented with a partially completed or incomplete PDI checklist and must identify missing required items based on standard automotive servicing procedures. This activity

Week	Learning Competency	Suggested Activities
		strengthens their ability to recognize essential PDI components and reinforces attention to detail. Analyze the incomplete PDI checklist and identify the missing required inspection items.
7	- Perform the pre-delivery inspection procedures. - Discuss the periodic maintenance services (PMS) for vehicle component parts (engine).	Hands-On PDI Practice - Learners perform a complete Pre-Delivery Inspection on a vehicle (or training model), checking engine fluids, brakes, tires, lights, safety features, and electrical systems. They practice following step-by-step PDI procedures while recording their findings. - Perform all PDI steps on the assigned vehicle using the standard checklist and document your inspection results. PDI Role-Play Service Simulation - Learners work in pairs or small groups simulating a real automotive service scenario where one acts as the service technician and the other as a customer. The technician performs the PDI procedures, explains each step, and ensures the vehicle meets pre-delivery standards.

Week	Learning Competency	Suggested Activities
		○ Simulate a vehicle PDI service, performing each inspection step and explaining your actions to the “customer.” ● Engine PMS Parts Matching Game ○ Learners are given cards or images of engine components (e.g., oil filter, spark plug, timing belt) and a list of periodic maintenance services (e.g., oil change, belt replacement, spark plug inspection). They must correctly match each engine part to its recommended PMS procedure and interval, reinforcing knowledge of maintenance schedules. ○ Match each engine component with its corresponding periodic maintenance service and explain why it is necessary. ● Engine PMS Case Study Discussion ○ Learners analyze a case study of a vehicle that has not undergone regular engine maintenance. They identify which PMS tasks were neglected, discuss possible consequences on engine performance, and suggest a maintenance plan to prevent future issues.

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		Review the engine maintenance case study, identify missed PMS tasks, and recommend a proper maintenance plan.
8	- Explore career and business opportunities for periodic maintenance services. - Discuss the latest trends and advanced technology for PMS.	Automotive Career & Business Research Project - Learners research different career paths (e.g., automotive technician, service manager, maintenance consultant) and business opportunities (e.g., car service centers, mobile oil change services, engine diagnostic shops) related to periodic maintenance services. They prepare a short presentation showing required skills, potential earnings, and industry trends. - Research and present a career or business opportunity in periodic vehicle maintenance, including roles, skills, and benefits. PMS Business Plan Simulation - Learners work in groups to design a mini-business plan for a periodic maintenance service, including services offered, target customers, pricing, and marketing strategies.

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		They simulate how their service would operate in a real automotive workshop environment. ○ Create a mini-business plan for a vehicle maintenance service and explain how it serves customer needs and generates income. ● Engine PMS Materials Scavenger Hunt ○ Learners are given a list of engine components and PMS tasks (e.g., oil change, filter replacement, spark plug inspection) and must identify all the required tools, materials, and consumables needed for each task, such as wrenches, oil, filters, and cleaning agents. ○ Using the engine PMS checklist, gather or list all materials and tools needed for each maintenance task. ● PMS Required Items Matching Exercise ○ Learners receive cards or images showing engine components and PMS items (e.g., engine oil, timing belt, diagnostic tool). They match each engine part to the corresponding maintenance item required for its PMS procedure, reinforcing proper preparation for servicing.

Week	Learning Competency	Suggested Activities
		Match each engine component with the correct PMS tool, material, or consumable required for its maintenance.
9	- Perform the procedure of periodic maintenance servicing for vehicle component parts (engine). - Explain the operating principles of vehicle components parts. - Discuss the periodic maintenance services (PMS) for the vehicle component parts (drive train, brake system, suspension system, and steering system).	Hands-On Engine PMS Workshop - Learners perform step-by-step periodic maintenance on engine components such as oil and filter replacement, spark plug inspection, air filter cleaning, and checking belts and hoses. This gives learners practical experience in following proper PMS procedures safely and accurately. - Perform the full engine PMS procedure on the assigned vehicle or training model using the standard checklist and record your work. PMS Procedure Simulation and Peer Demonstration - Learners work in pairs or small groups to simulate engine PMS tasks. One student performs the procedure while explaining each step aloud, and peers observe to ensure accuracy and safety. Afterward, roles are switched so each learner practices and teaches. - Simulate and perform engine PMS

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		tasks while explaining each step to peers, ensuring all procedures are followed correctly. • Component Function Demonstration ◦ Learners select a vehicle component (e.g., engine, transmission, braking system, alternator) and demonstrate its function using a physical model, diagram, or actual vehicle. They explain how the component works, its role in vehicle operation, and how it interacts with other systems. ◦ Demonstrate a selected vehicle component and explain its operating principle and role in the vehicle system. • How It Works - Group Diagram Activity ◦ Learners work in groups to create a labeled diagram or flowchart showing a vehicle component's parts and how they function together. They present their diagram and explain the operating principles step by step to the class. ◦ Draw a diagram of a vehicle component and explain how its parts operate together to perform its function.

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		• PMS Systems Analysis Workshop ○ Learners examine the drive train, brake, suspension, and steering systems of a vehicle (or training model) and identify the required periodic maintenance tasks for each system, such as fluid replacement, lubrication, belt inspection, brake pad checks, and alignment checks. They then explain the importance of each PMS task in maintaining vehicle performance and safety. ○ Inspect each vehicle system and discuss the periodic maintenance tasks required for safe and efficient operation. • PMS Case Study Discussion ○ Learners analyze a case study of a vehicle that has experienced problems due to neglected PMS of the drive train, brakes, suspension, or steering system. They identify which maintenance services were missed, discuss the consequences, and suggest proper PMS schedules for each system. ○ Review the case study, identify neglected PMS tasks in each system,

Week	Learning Competency	Suggested Activities
		and explain how proper maintenance could prevent issues.
10	Identify required items for periodic maintenance service on the vehicle.	PMS Tools and Materials Inventory - Learners are provided with a list of PMS tasks (oil change, filter replacement, brake inspection, tire rotation, etc.) and must identify all the required tools, consumables, and safety equipment needed to perform each task correctly. - List or gather all tools, materials, and safety equipment required for each periodic maintenance task on a vehicle. PMS “Missing Items” Challenge - Learners are presented with a PMS setup that is intentionally missing some tools or materials. They must identify what items are missing for a specific maintenance task and explain why each item is necessary. - Examine the PMS setup, identify missing tools or materials, and explain their importance for proper maintenance. Hands-On Vehicle Systems PMS Workshop

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		○ Learners perform periodic maintenance tasks on the drive train, brake system, suspension, and steering system of a vehicle (or training model). Tasks include checking and topping up fluids, inspecting brake pads, lubricating joints, checking tire alignment, and tightening steering components, giving learners practical experience in maintaining these key systems. ○ Perform all required PMS tasks on the assigned vehicle systems following the standard procedure and record your observations. ● PMS Peer Demonstration and Simulation ○ Learners work in pairs or small groups to simulate PMS procedures on each vehicle system. One student performs the maintenance steps while explaining the procedure and purpose to peers, then they switch roles so everyone gains hands-on practice and understanding. ○ Simulate and perform PMS procedures on vehicle systems while explaining each step to peers to ensure accuracy and understanding.

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11	- Discuss vehicle releasing procedures of the vehicle. - Calculate service cost for PMS.	Vehicle Releasing Procedure Walkthrough - Learners review the standard steps for releasing a vehicle to a customer after servicing, including verifying completed work, checking inspection reports, explaining maintenance performed, obtaining customer acknowledgment, and providing safety reminders. - Follow and explain each step of the vehicle releasing procedure using a sample checklist or case scenario. Role-Play Customer Vehicle Release - Learners simulate a real service scenario where one student acts as the service technician and another as the customer. The technician demonstrates the proper vehicle releasing procedure, communicates completed maintenance, explains safety tips, and answers customer questions. - Simulate releasing a serviced vehicle to a customer while explaining all completed procedures and safety instructions. PMS Service Cost Calculation Drill - Learners are given a list of PMS tasks

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		(e.g., oil change, filter replacement, brake inspection, tire rotation) along with the cost of parts, materials, and labor. They calculate the total service cost for completing all tasks on a vehicle, reinforcing practical math skills in an automotive context. ○ Calculate the total cost of PMS for a vehicle by adding up the cost of materials, parts, and labor for each maintenance task. ● PMS Service Quotation Simulation ○ Learners simulate preparing a service quotation for a customer. They select required PMS tasks, determine material and labor costs, apply any discounts or taxes, and present a professional service cost estimate. ○ Prepare a detailed PMS service quotation for a customer, showing all materials, labor costs, and the total amount payable.
Suggested Performance Tasks	● Individual Performance Task - Vehicle Lifting and Positioning Drill ○ The learner individually demonstrates proper use of vehicle lifting equipment (like hydraulic lifts or jacks), positions a vehicle safely in the service bay, and ensures correct placement of support stands, following all safety procedures. ○ Individually lift, position, and secure a vehicle using lifting and support equipment	

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		while observing all safety precautions. Group Performance Task - Team-Based Vehicle Maneuver and Safety Simulation - Learners work in small groups to safely move and park a light vehicle in a simulated workshop environment. The team plans the maneuver, guides the driver, ensures proper communication, and positions the vehicle accurately, applying safety protocols throughout. - As a team, safely maneuver and park a vehicle in the designated area while guiding the driver and observing all safety measures. Individual Performance Task - Pre-Delivery Inspection (PDI) Checklist Exercise - The learner performs a complete PDI on a vehicle individually, inspecting key systems such as engine, brakes, lights, tires, fluids, and safety features. The learner documents findings, identifies any issues, and applies safety precautions throughout the process. - Individually perform a full PDI on a vehicle using the checklist while following all safety precautions and recording your observations. Group Performance Task - PDI Team Inspection Simulation - Learners work in small groups to inspect a vehicle collaboratively. Each member focuses on specific systems (engine, brakes, electrical, suspension), and the team discusses findings, summarizes overall vehicle readiness, and ensures all safety measures are observed. - As a group, conduct a PDI on a vehicle, inspect assigned systems, share findings, and ensure safety procedures are followed. Individual Performance Task - Engine PMS Hands-On Drill - The learner performs periodic maintenance tasks on an engine individually, such as oil and filter replacement, spark plug inspection, checking belts and hoses, and topping up fluids, while observing all safety procedures. - Individually perform PMS tasks on an engine following the proper procedure and safety precautions, and record your results.

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		• Group Performance Task - Engine PMS Collaborative Workshop ○ Learners work in small groups to conduct a complete engine PMS on a training vehicle. Each member is assigned a specific task (oil change, fluid check, belt inspection, spark plug replacement), and the team ensures the procedure is completed accurately while following safety standards. ○ As a group, perform engine PMS tasks collaboratively, with each member completing assigned tasks while observing safety procedures. • Individual Performance Task - Comprehensive Vehicle PMS and Cost Calculation ○ The learner individually performs periodic maintenance on key vehicle systems—drive train, brake system, suspension, and steering—ensuring all procedures are done safely. After completing the maintenance, the learner calculates the total service cost and prepares the vehicle for release. ○ Individually perform PMS on vehicle systems, calculate total service cost, and prepare the vehicle for release while following safety precautions. • Group Performance Task - Team-Based Vehicle PMS Simulation and Release ○ Learners work in small groups to complete a full PMS on a training vehicle. Each member is assigned specific systems (e.g., one handles brakes, another suspension, another drive train/steering). After completing the maintenance, the team calculates the total service cost collaboratively and simulates the proper vehicle release procedure to a “customer.” ○ As a team, perform PMS on assigned vehicle systems, calculate total service cost, and simulate releasing the vehicle safely to the customer.